

Lesson 05

Soil Compaction



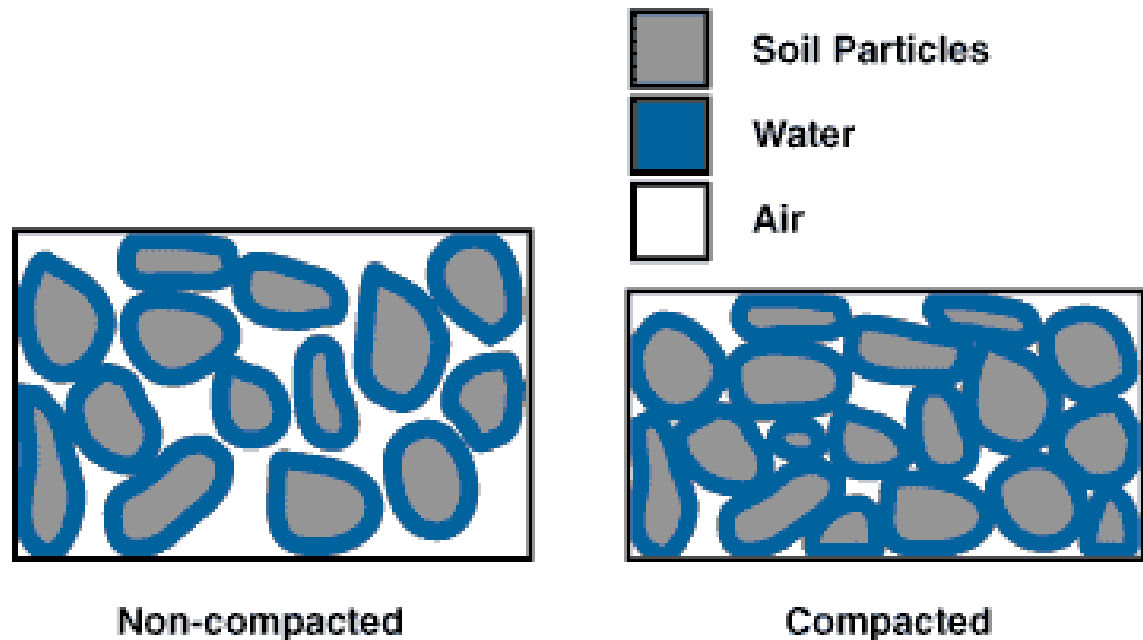
Topics to be covered;

1. What is soil compaction?
2. Parameters affecting compaction
3. Methods of Testing Compaction of Soil
 - Laboratory tests
 - Field (insitu) tests

What is soil compaction?

Soil compaction is the method of **mechanically** increasing the density of a soil by packing the soil particles closer together.

Packing of the soil particles closer together is done by the removal of air from the soil voids by tamping, vibrating or rolling (ie. using mechanical energy) resulting in the reduction of the soil volume and increase in density.



Soil is compacted to improve the engineering properties of the soil.

- Strength
- Compressibility
- permeability



Why Compact?

Soils are compacted to:

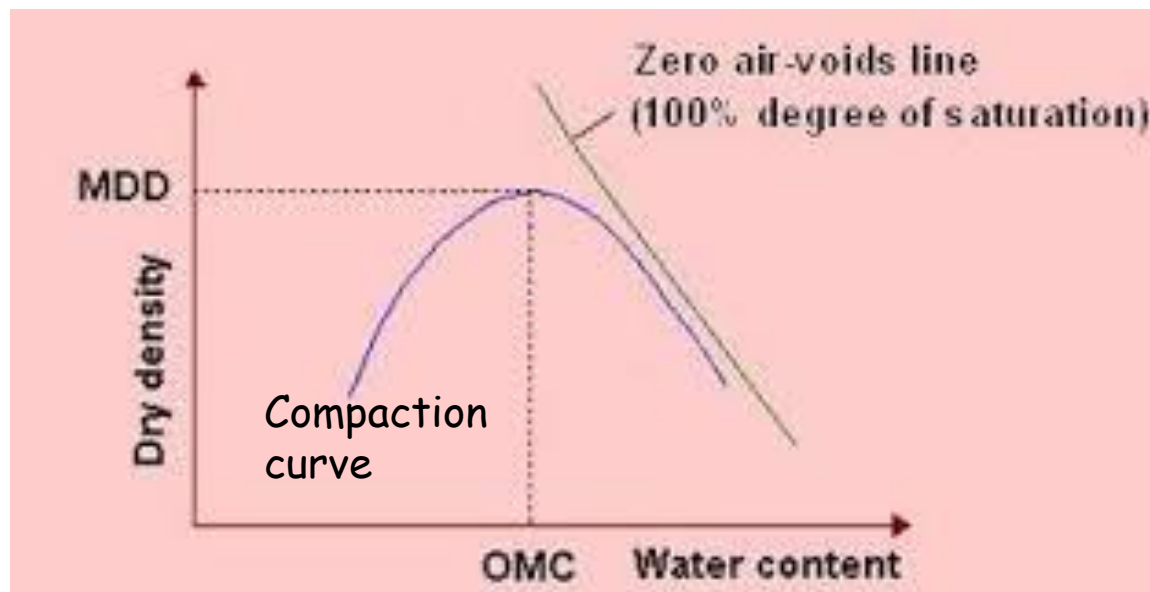
- Increase load-bearing capacity/shear **strength** of soil
- Improve stability of soil and reduce future **settlement**
- Reduce voids ratio and reduce soil **permeability**

Measurement of Compaction

Compaction is measured using the **dry density** of the compacted soil.

Compaction curve

The plot of dry density against the water content is known as the **compaction curve**.

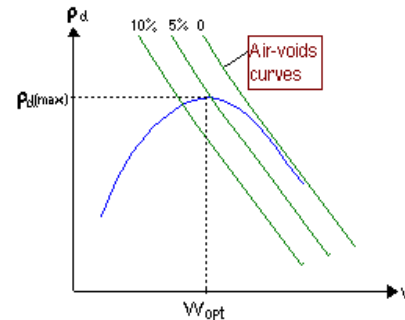


Zero Air Voids Line

zero air voids is a theoretical limit which can never be reached in practice, it is impossible to fully expel the void air in the compaction process.

Parameters affecting soil compaction:

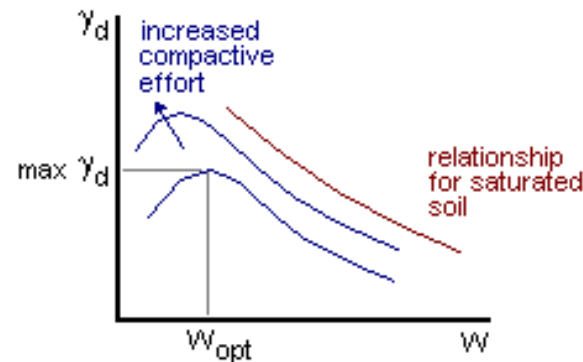
- Soil moisture content



- Soil type (sand/clay, grading, plasticity)



- Compactive effort (type of plant, weight, no of passes)



Soil Moisture Content

Moisture content of the soil is vital to proper compaction.

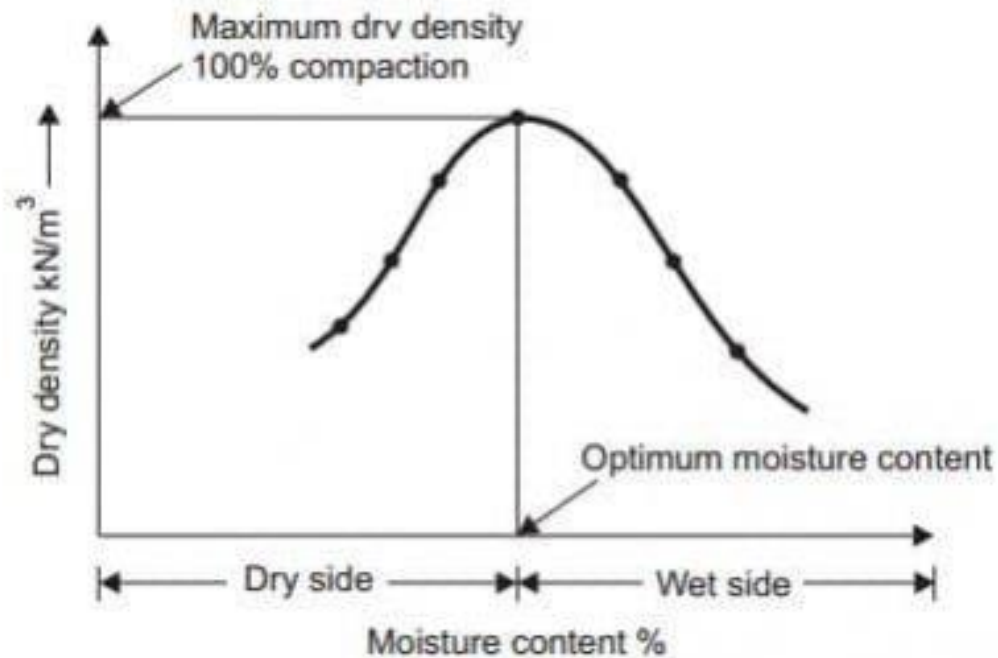
When water is added to a dry soil;

- Air will be forced out from soil voids
- Water will start to lubricate the soil solids

and soil will pack in to a denser structure. This will result in the **increase in the dry density** of soil.

When most of the air is removed from the voids, addition of more water results in pushing the soil solids away from each other with an increase in the volume. This will result in **decreasing the dry density** of soil.

Variation in compaction with water content

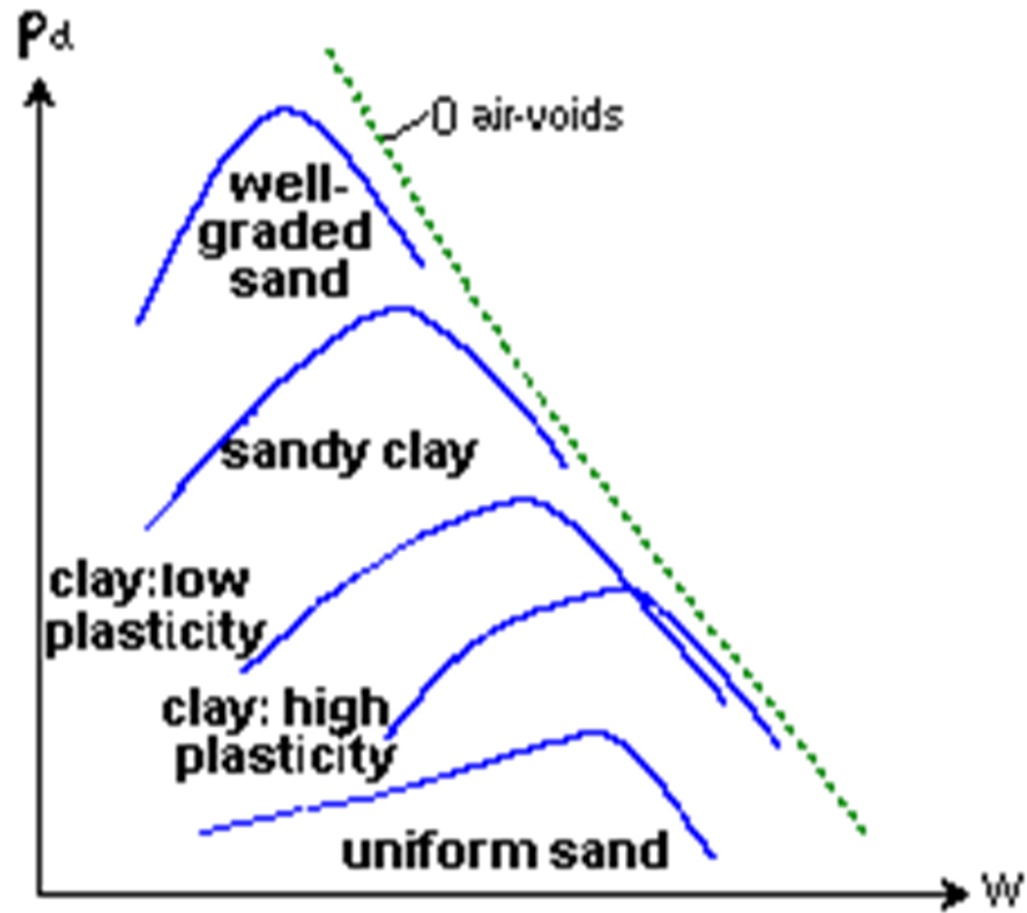


The maximum dry density for most soils is at a certain water content (OMC) for a given compactive effort.

Soil Type

Soils have different properties due to variations in size distribution, shape and specific gravity of particles and due to amount and type of clay minerals present.

The soil type has a great influence on the maximum dry density and optimum moisture content.



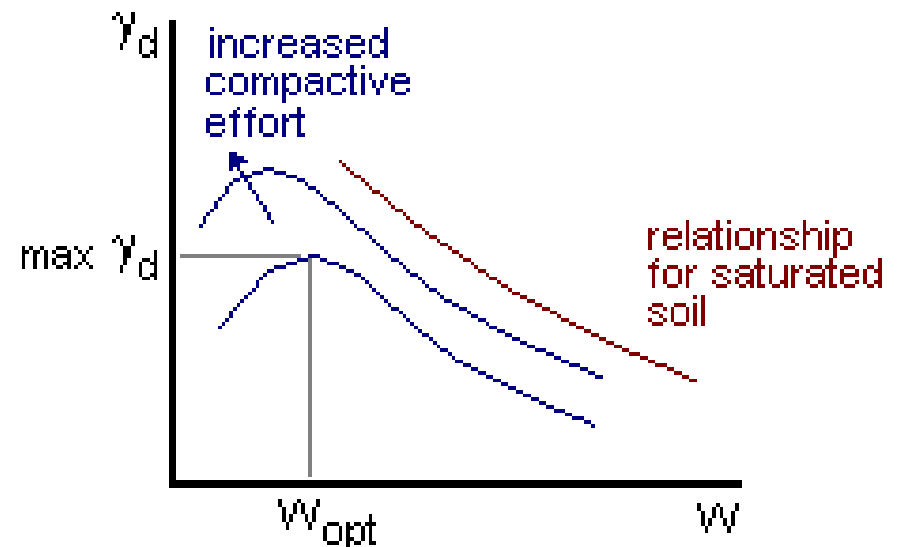
The Compactive Effort

Compaction of soil is accomplished by use of heavy machines/equipment.

The compactive effort of heavy machines can be varied by;

- adjusting the weight of the machine
- adjusting the number of passes applied

As the compactive effort is increased the maximum dry density is increased.



Methods of Testing Compaction of Soil

- Laboratory tests
- Field/in situ tests

Before field compaction it is necessary to assess the amount of compaction of soil and the water content required in the field; hence compaction tests are done in the laboratory.

After compaction, it is necessary to determine if the compacted soil density meets specifications. This can be done on site.

Laboratory Compaction Tests

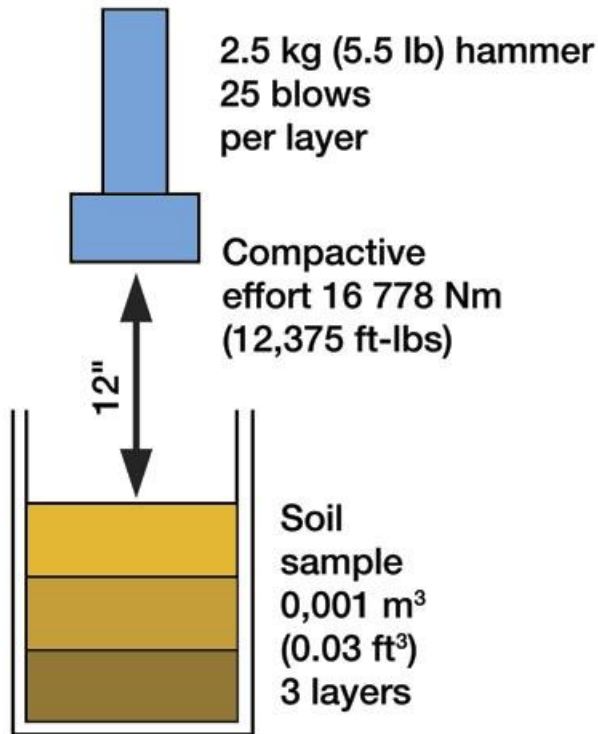
The most commonly used laboratory test is called the Proctor compaction test

This test determines the optimum moisture content at which a given soil type will achieve its maximum dry density.

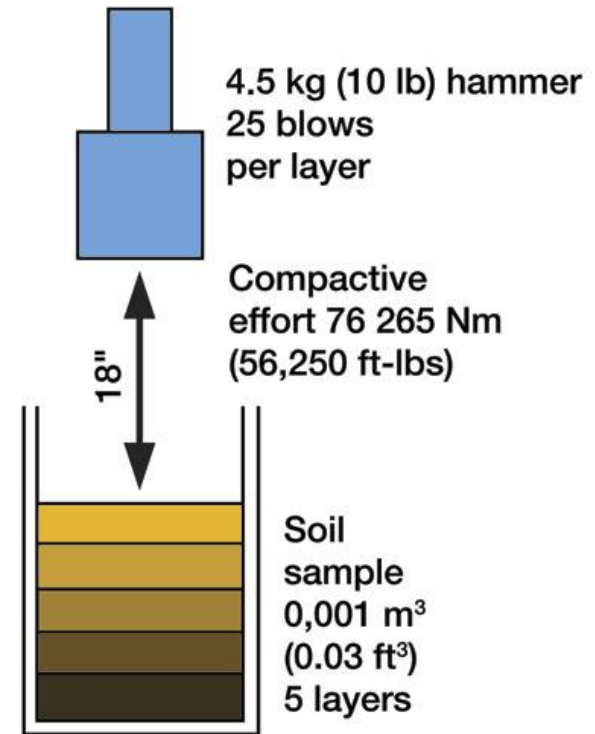
The term Proctor is in honor of R. R. Proctor, who in 1933 showed that the dry density of a soil for a given compactive effort depends on the amount of water the soil contains during soil compaction.

The Proctor compaction test can be carried out in two ways.

Standard Proctor Test



Modified Proctor Test



The modified test is used in testing soil needed to support heavier loads such as a busy highways or air strips

Standard Proctor Test

Procedure

- Mix the soil with some water
- Weigh the empty mould
- compact the moist soil in to the mould in 03 layers
- Compact each layer with a standard Compactive effort
- Weigh the mould with sample
- Determine the water content of soil
- Repeat the steps increasing the water content

Apparatus



Calculations

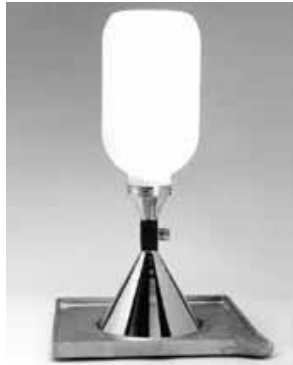
1. For each compacted specimen determine;
 - the bulk density
 - the moisture content
 - the dry density
2. Draw the compaction curve
3. Determine the optimum moisture content and max dry density

In-situ Density Tests

After compaction, it is necessary to determine if the compacted soil density meets specifications. This can be done on site using two basic tests.

1. Sampling tests

Sand cone test



2. Nuclear tests



Core cutter test



Sand Cone Test

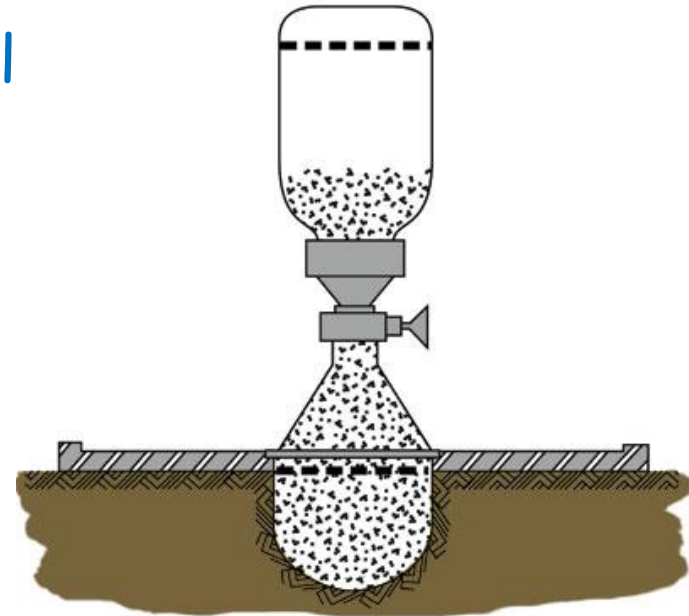
Procedure

- Dig a test hole in the compacted soil
- Remove the soil from the hole
- Weigh the soil sample
- The hole is filled with dry sand from a graduated bottle.
- The sand has a known density.
- Determine the water content of soil

Calculations

Determine ;

- the bulk density
- moisture content
- dry density of soil.



Core Cutter test

Procedure

- The cutter is pressed into the soil with the help of a dolly and rammer so that it is filled with the soil without disturbing the core contents.
- The cutter filled with the soil is lifted up.
- The mass of the soil in the cutter is determined.
- The water content of the sample is determined.

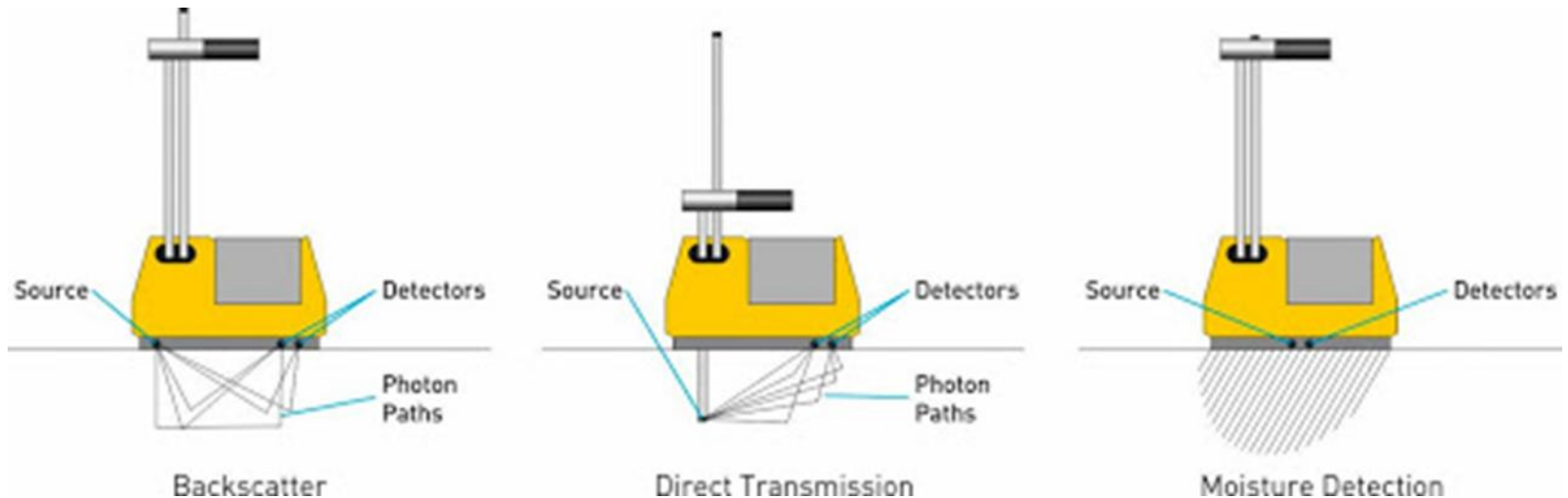


Using the values of bulk density and the water content the dry density of the soil can be calculated.

Nuclear Tests

Nuclear Density meters are a quick and fairly accurate way of determining density and moisture content of a compacted soil.

The meter uses a radioactive source (Cesium 137) at the soil surface (backscatter) or from a probe placed into the soil (direct transmission).



Summary;

1. Soil compaction is the removal of air from soil voids using mechanical energy.
2. Soil compaction will improve the strength & stability of soil while reducing its permeability.
3. Soil compaction is affected by; soil moisture, soil type and the compactive effort.
4. Compaction can be measured by laboratory and insitu testing.